UIT
Dept. of Computer Network

ARRIVAL PROCESSES & QUEUING SYSTEMS

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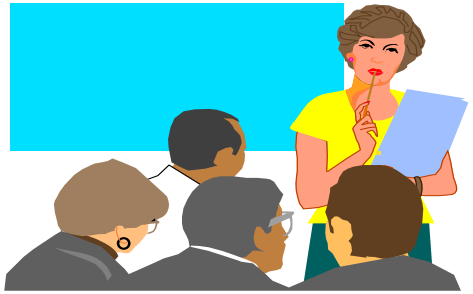

Feb.2016

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1. Classification of queuing models

D. G. Kendall introduced the following notation for queuing models:

A/B/n

Where: A (Arrival Process), B (Service process), n (number of servers)

The following standard notations are often used:

M ~ Markovian. Exponential time intervals (Poisson arrival process, exponentially distributed service time)

D ~ Deterministic. Constant time interval

E_k ~ Erlang-k distributed time interval (E_1 = M)

G ~ General. Arbitrary time distribution

GI ~ General Independent

GEO ~ Geometric distribution

Examples

M/M/n is a pure delay system with Poisson arrival process, exponentially distributed service times and n servers. It is classical Erlang delay system

GI/G/1 is a general delay system with only one server

1. Classification of queuing models

Extended notations

The above notation is used everywhere in literature. For a complete specification of a queuing system, more information is required:

A/B/n/K/N/X

Where: K (total capacity of the system), N (number of customers), X (queuing discipline)

K = n corresponds to a loss system, which is also denoted as **A/B/n-Loss**

Note that **Full Availability** is usually assumed

Queuing Disciplines

Customers waiting in a queue to be served can be selected for service according to many different principles

FIFO First In – First Out. It is often called a fair queue or an ordered queue. It is also denoted as FCFS: First Come – First Served

LIFO Last In – First Out. This is the stack principle. It is used e.g. in storages, on shelves of shops etc. This discipline is also denoted as LCFS

1. Classification of queuing models

SIRO Service In Random Order. All customers waiting in the queue have the same probability of being chosen for service. This is also called RANDOM or RS (Random Selection).

The total waiting time for all customers is the same in the above three systems. The queuing discipline only decides how the total waiting time is allocated to the customers.

The following disciplines try to reduce the total waiting time. The service time is used as criterion:
SJF Shortest Job First (SJN = Shortest Job Next, SPF = Shortest Processing Time First, etc.). This discipline assumes that the service time is known in advance. This queuing discipline minimizes the total waiting time for all customers.

1. Classification of queuing models

The following disciplines take account of both the arrival times and service times:

RR Round Robin. A customer served is given at most a fixed service time (time slice). If the service time is not completed during this interval, the customer returns to the queue which is FIFO

PS Processor Sharing. All customers share the service capacity equally

Priority of customers

Customers are divided into p priority classes, where a customer, who belongs to class k has higher priority than a customer, who belongs to class k+1.

NON-PREEMPTIVE = HOL (Head-of-the-Line): An arriving customer with higher priority than a customer being served waits until a server becomes idle (and all higher priority customers have been served)

PREEMPTIVE: A customer being served having lower priority than the arriving customer is interrupted. We distinguish between:

1. Classification of queuing models

Preemptive resume, where the service is continued from, where it was interrupted

Preemptive without resampling, where the service restarts from the beginning with the same service time

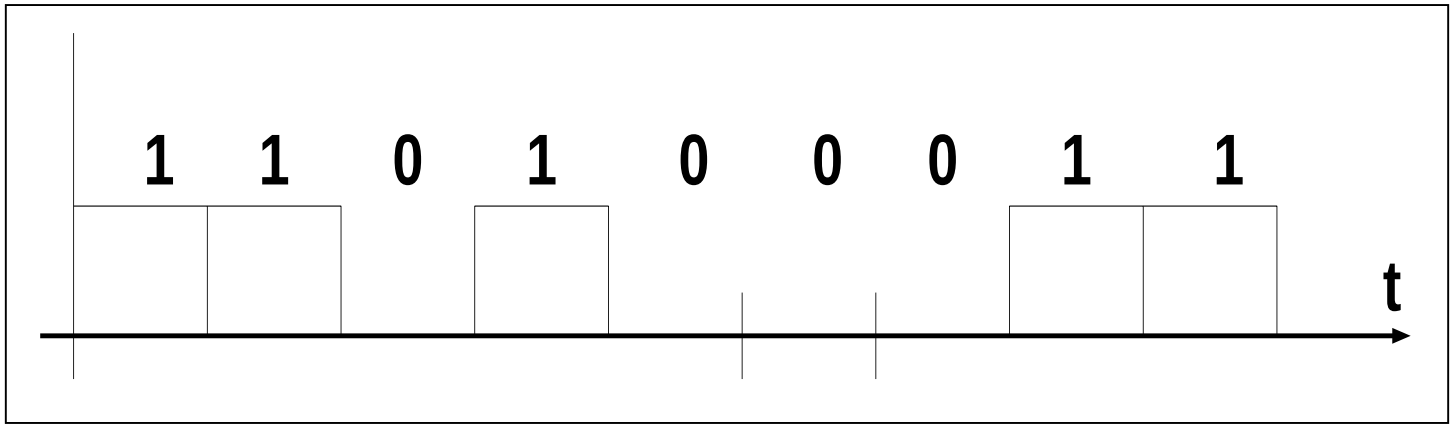
Preemptive with resampling, where the service starts again with a new sampled service time

GD General Discipline. An arbitrary queuing discipline

2.3 Bernoulli Model

p=P{arrival in slot}

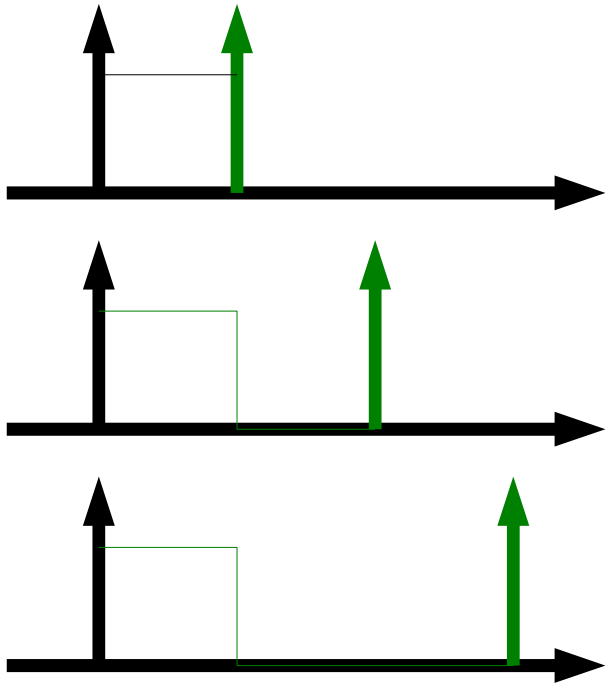
1-p=P{no arrival in slot}



2.4 Geometric Model

p=P{arrival in slot}

1-p=P{no arrival in slot}



$$t = 1 \rightarrow p_1 = p$$

$$t = 2 \rightarrow p_2 = (1-p)p$$

$$t = 3 \rightarrow p_3 = (1-p)^2 p$$

$$\cdots\cdots$$

$$t = n \rightarrow p_n = (1-p)^n p$$

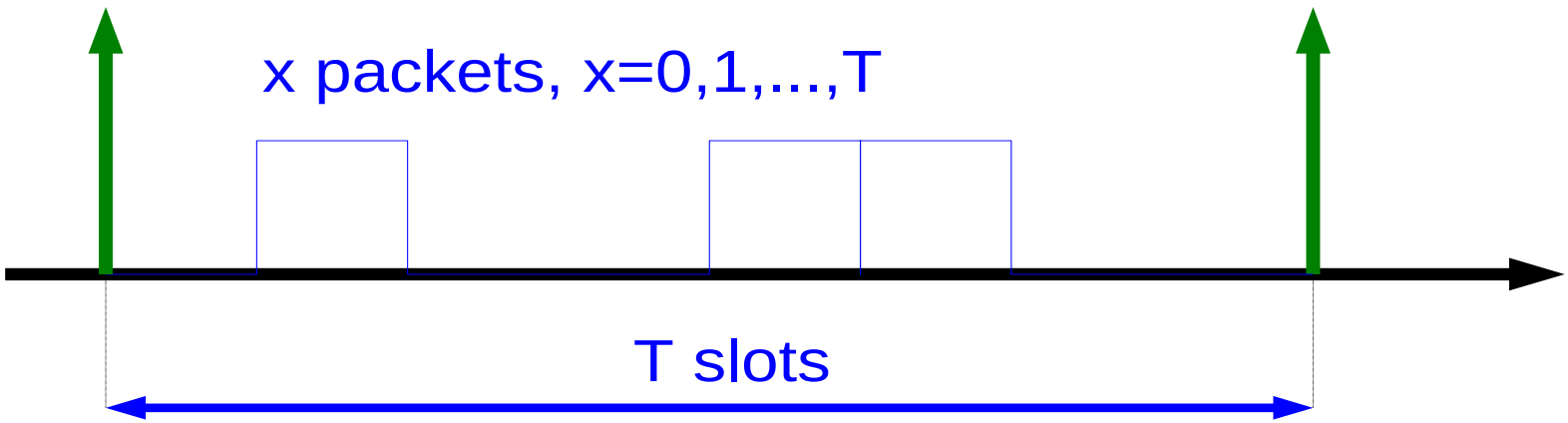
Pn=P{1 arrival in next n time slots}

2.5 Binomial Model

T independent slots

p=P{arrival in one slot}

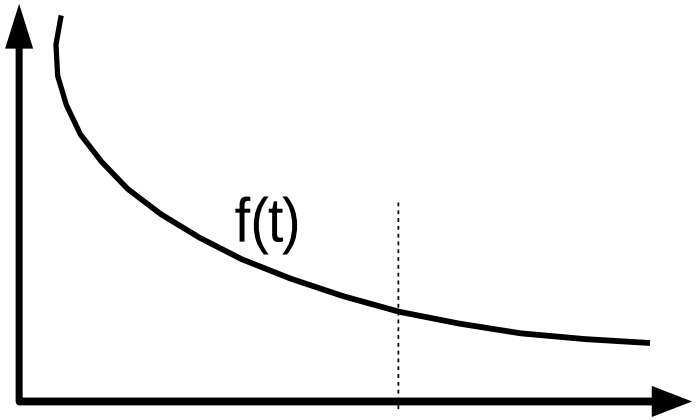
1-p=P{no arrival in one slot}

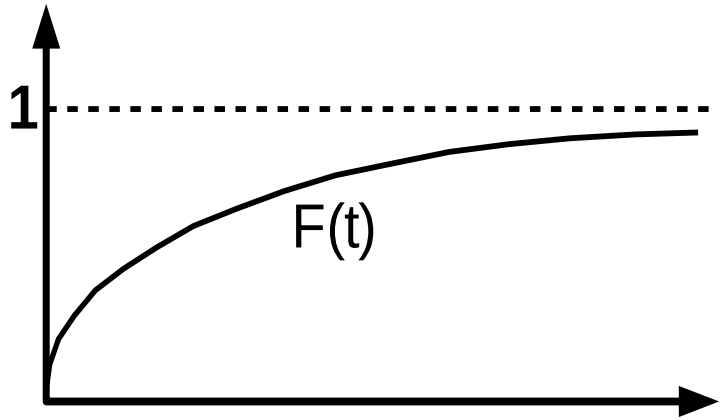


Binomial probability distribution :

$$P_x = \binom{T}{x} p^x (1-p)^{T-x}$$

2.1 Exponential Model





✓Probability density function (pdf): $f(t) = \lambda \cdot e^{-\lambda t}$

✓Cumulative function: $F(t) = 1-e^{-\lambda t}$ = Probability{inter-arrival time ≤ t}.

✓This kind of system has no history because probability to get a new arrival is independent of the previous arrival. The value of t has no influence. This is called as **memoryless** system (**Markov** system).

✓The **memoryless** property says that one can predict the future state of the system based on the present state without knowing the past history of the state of the system.

2.1 Exponential Model

✓Mean value: $E[x] = 1/\lambda$

✓Variance: $Var[x] = 1/\lambda^2$

2.1 Exponential Model

Proof of memoryless property:

Conditional Probability (P) to get next arrival during (t, t+dt) when we know that it did not come before t is:

$$P = \frac{P(t < t' \leq t + dt)}{1 - P(t' \leq t)} = \frac{F(t + dt) - F(t)}{1 - F(t)}$$

$$P = \frac{(1 - e^{-\lambda (t+dt)}) - (1 - e^{-\lambda \mathrm{t}})}{1 - (1 - e^{-\lambda \mathrm{t}})}$$

$$P = 1 - e^{-\lambda \mathrm{d}t} = F(dt)$$

Microsoft Excel - arrival_process_exp.xls

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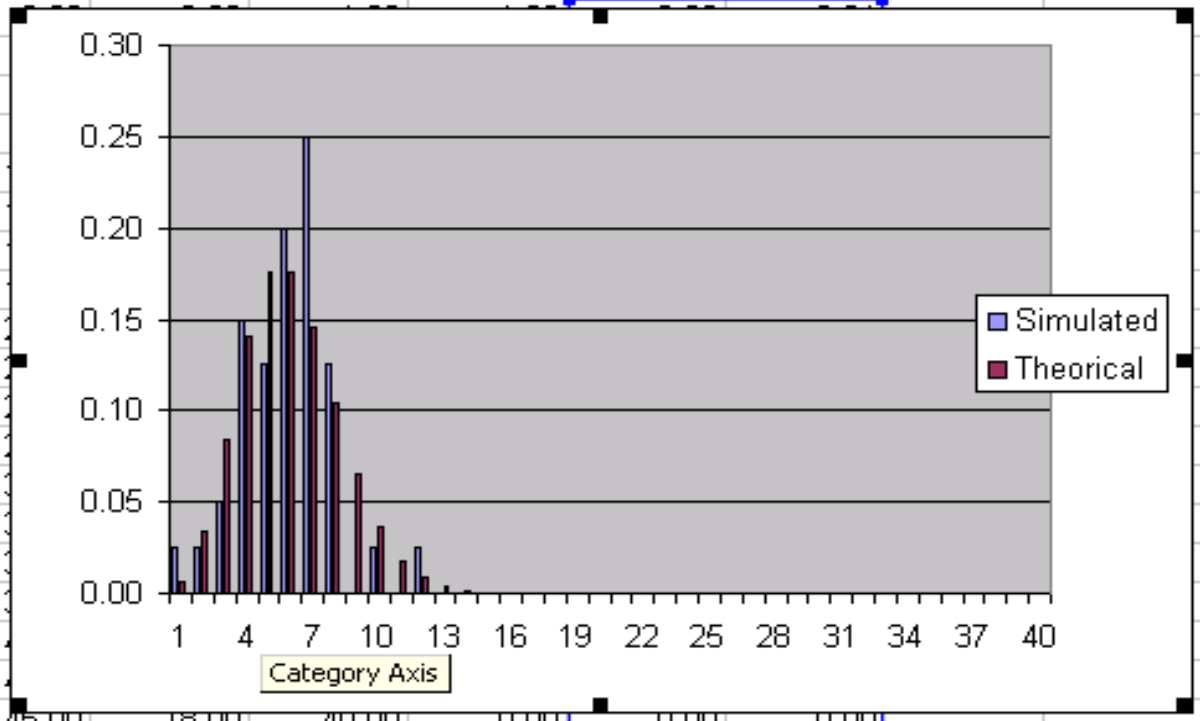
Arial 10

Chart Area

	A	B	C	D	E	F	G	H	I	J	K	L
1	Lambda	2.00		2.5		Number	intervals	40.00		No. of arr.	<=Ts	
2		Mean t(i)		|	Var	4.35641	Mean	4.95	<=====	No. of arr.	in Ts	
3		0.503259		|				Sum Prob.	Sum Prob.			
4	Index	inter-arr.	arr.-time	interval				1.00	1.00			
5	i	t(i)	t	Ts	x	Poiss. F(x)	Poiss. f(x)	%real	%theory	Interval	Ts	2.50
6	1.00	0.80482	0.80482									
7	2.00	0.26098	1.0658									
8	3.00	0.988875	2.054675									
9	4.00	0.099118	2.153793									
10	5.00	0.14647	2.300263									
11	6.00	0.047639	2.347902									
12	7.00	0.682329	3.030231									
13	8.00	0.15015	3.180381									
14	9.00	0.86073	4.041111									
15	10.00	0.092248	4.13336									
16	11.00	0.883923	5.017283									
17	12.00	0.719678	5.736961									
18	13.00	0.932517	6.669478									
19	14.00	0.014518	6.683996									
20	15.00	0.155559	6.839554									
21	16.00	0.037152	6.876706									
22	17.00	0.348615	7.225322									
23	18.00	0.642429	7.867751									
24	19.00	0.923418	8.79117									



Sheet1 Sheet2 Sheet3

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2.2 Poisson Model

(Number of arrivals during observation period t)

✓Interval approach -> time measurement durations

✓Number approach -> counting method

With exponential distributed intervals (λ = mean arrival rate), and observation period T:

➢Expected number of arrivals in T : λT

➢Probability to get x arrivals in T is **Poisson distributed**

$$P_x(T) = \frac{(\lambda . T)^x}{x!} . e^{-\lambda . T}$$

$$Mean\,value : E[x] = \lambda T$$

$$\text{Variance} : \text{Var}[x] = \lambda T$$

Proof

Suppose the time period (0,t) is divided into a large number (n) of small time intervals of width Δt. If intervals are very small, there will be one arrival or no arrival within any Δt. The intervals will behave as a Binomial events having either value "1" or "0". We will let the probability of a single arrival in an interval be proportional to the length of the interval Δt

	1 0	1 0		1 0	
0	dt 1	dt 2	----------	dt n	**t**

n = t/Δt

P{exactly 1 arrival in [t,t+Δt]} = λ.Δt, where λ = mean arrival rate

P{no arrival in [t,t+Δt]} = 1 – λ.Δt

This is the **Binomial distribution**, therefore Probability of x arrivals during the period [0,t] with particular order is:

$$P_x(\lambda t) = \frac{(\lambda t)(\lambda t)\ldots(\lambda t)}{x-terms} \cdot \frac{(1-\lambda t)(1-\lambda t)\ldots(1-\lambda t)}{(n-x)-terms}$$

$$P_x(\lambda t) = (\lambda t)^x.(1-\lambda t)^{n-x}$$

But these arrivals can occur in different ways

Therefore the probability of x arrivals during the interval [0,t] is binomially distributed:

$$P_x(\lambda\, t) = \binom{n}{x}.(\lambda\, \Delta t)^x.(1-\lambda\, \Delta t)^{n-x} = \binom{n}{x}.(\lambda\, \frac{\mathrm{t}}{\mathrm{n}})^x.(1-\lambda\, \frac{\mathrm{t}}{\mathrm{n}})^{n-x}$$

$$P_x(\lambda\, t) = \frac{n!}{x!(n-x)!}(\lambda\, \frac{\mathrm{t}}{\mathrm{n}})^x.(1-\lambda\, \frac{\mathrm{t}}{\mathrm{n}})^{n-x}$$

$$as\, n \to \infty, \Delta t \to 0\, then$$

$$P_x(\lambda\, t) = \frac{(\lambda\, t)^x}{x!}.Lim_{n\to\infty}\frac{n!}{(n-x)!}.\frac{(1-\frac{\lambda\, t}{n})^{n-x}}{n^x}$$

$$P_x(\lambda\, t) = \frac{(\lambda\, t)^x}{x!}.Lim_{n\to\infty}\frac{n(n-1)\ldots(n-x+1)}{n.n\ldots n}.(1-\frac{\lambda\, t}{n})^n.(1-\frac{\lambda\, t}{n})^{-x}$$

$$P_x(\lambda\, t) = \frac{(\lambda\, t)^x}{x!}.Lim_{n\to\infty}(1-\frac{\lambda\, t}{n})^n = \frac{(\lambda\, t)^x}{x!}.e^{-\lambda\, t}\ \text{Poisson dist.}$$

2.2 Poisson Model

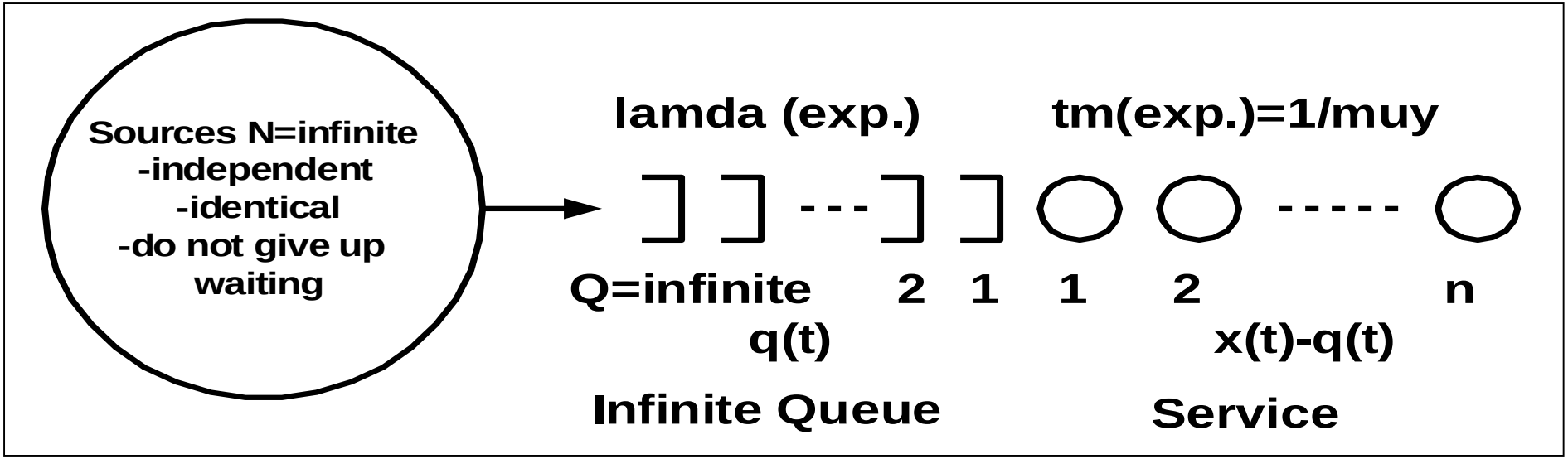
Theorem

If the inter-arrival times has exponential distribution, then the number of arrivals in any duration T has Poisson distribution.

3. Erlang's Delay System (M/M/n - System)

Assumptions

- ✓Limited number of servers ($n<\infty$)
- ✓Infinite number of queuing positions ($Q=\infty$)
- ✓Inter-arrival times and service times exponentially distributed
- ✓Sources are identical and independent of each others
- ✓Customers do not give up in the queue, therefore all requests are served
- ✓FIFO queue discipline



3. Erlang's Delay System (M/M/n - System)

✓The inter-arrival time distribution is exponential with mean arrival rate **λ**

✓The probability distribution of the duration of an individual occupation is exponential with mean t_m, $\mathbf{t_m = 1/\mu}$, **μ** is the mean service (departure) rate

✓**A**, number of arrivals within t_m → $\mathbf{A = \lambda . t_m}$

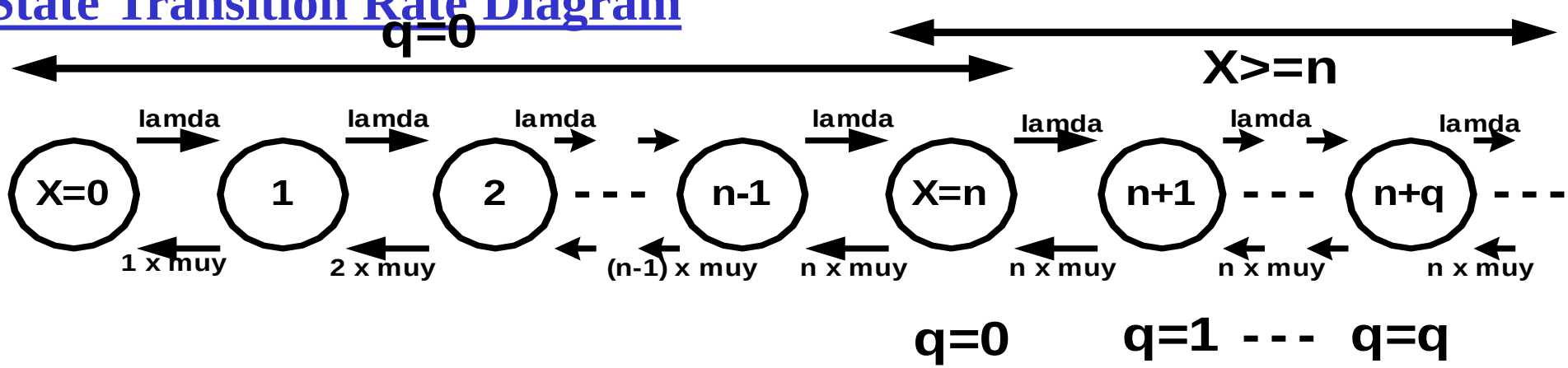
✓Full availability, i.e, any arrival can find and use any idle servers

x(t) = number of customers in the system at time t = (0,1,2,…,n,n+1,…,∞)

q(t)= number of customers in the queue at time t = (0,1,2,…,∞)

x(t)-q(t) = number of customers in service at time t = (0,1,2,…,∞)

State Transition Rate Diagram



3. Erlang's Delay System (M/M/n - System)

Cut - equations between adjacent states (Equilibri um state) :

$$\lambda.P_0 = \frac{1}{t_m}.P_1 \rightarrow P_1 = (\lambda.t_m).P_0 = A.P_0$$

$$\lambda.P_1 = \frac{2}{t_m}.P_2 \rightarrow P_2 = \frac{(\lambda.t_m)^2}{2!}.P_0 = \frac{A^2}{2!}.P_0$$

$$\lambda.P_{n-1} = \frac{n}{t_m}.P_n \rightarrow P_n = \frac{(\lambda.t_m)^n}{n!}.P_0 = \frac{A^n}{n!}.P_0$$

$$\lambda.P_n = \frac{n}{t_m}.P_{n+1} \rightarrow P_{n+1} = \frac{\lambda.t_m}{n}.P_n = \frac{A}{n}.P_n$$

$$\lambda.P_{n+1} = \frac{n}{t_m}.P_{n+2} \rightarrow P_{n+2} = \frac{\lambda.t_m}{n}.P_{n+1} = \left(\frac{A}{n}\right)^2.P_n$$

$$Therefore, P_{n+q} = \left(\frac{A}{n}\right)^q.P_n$$

$$\rightarrow P_x = \left\{ \begin{array}{l} \frac{A^x}{x!}.P_0 \;\; for \;\; x = 0,1,\ldots,n-1 \\ \left(\frac{A}{n}\right)^q.\frac{A^n}{n!}.P_0 \;\; for \;\; x \geq n, x = n+q \end{array} \right\}$$

3. Erlang's Delay System (M/M/n - System)

Normalization :

$$\sum_{x=0}^{\infty} P_x = 1 = \sum_{x=0}^{n-1} P_x + \sum_{q=0}^{\infty} P_{n+q} = \sum_{x=0}^{n-1} \frac{A^x}{x!}.P_0 + \sum_{q=0}^{\infty} \left(\frac{A}{n}\right)^q . \frac{A^n}{n!}.P_0$$

$$\sum_{q=0}^{\infty} \left(\frac{A}{n}\right)^q = \frac{1}{1-\frac{A}{n}} = \frac{n}{n-A} \quad if \quad \frac{A}{n} < 1$$

$$\rightarrow P_0 = \frac{1}{\sum_{i=0}^{n-1} \frac{A^i}{i!} + \frac{n}{n-A} . \frac{A^n}{n!}}$$

$$\rightarrow P_x = \left\{ \begin{array}{l} \dfrac{\frac{A^x}{x!}}{\sum_{i=0}^{n-1} \frac{A^i}{i!} + \frac{n}{n-A} . \frac{A^n}{n!}} \quad where\ q = 0, x = 0,1,\ldots,n-1 \\ \dfrac{\left(\frac{A}{n}\right)^q . \frac{A^n}{n!}}{\sum_{i=0}^{n-1} \frac{A^i}{i!} + \frac{n}{n-A} . \frac{A^n}{n!}} \quad where\ x \geq n, and\ \ q = x - n = 0,1,\ldots,\infty \end{array} \right\}$$

3. Erlang's Delay System (M/M/n - System)

$\mathbf{P_{n+q}}$ gives the probability that there are exactly q customers in the queue, and all servers are busy.

The probability that an arrival has to enter the queue = P{an arrival to be delayed}, is given by **Erlang's Delay Formula**, (or Erlang's Second Formula, or Erlang's c-Formula):

P{an arrival to be served immediately} = 1 – $\mathbf{D_n(A)}$

$$D_n(A) = \sum_{x=n}^{\infty} P_x = \sum_{q=0}^{\infty} P_{n+q} = \sum_{q=0}^{\infty} \frac{\left(\frac{A}{n}\right)^q \cdot \frac{A^n}{n!}}{\sum_{i=0}^{n-1} \frac{A^i}{i!} + \frac{n}{n-A} \cdot \frac{A^n}{n!}}$$

$$D_n(A) = \frac{\frac{n}{n-A} \cdot \frac{A^n}{n!}}{\sum_{i=0}^{n-1} \frac{A^i}{i!} + \frac{n}{n-A} \cdot \frac{A^n}{n!}}$$

At least q arrivals waiting

P{at least q arrivals waiting} = P{x≥n+q}

$$P_{x\geq n+q} = \sum_{x=n+q}^{\infty} P_x = \frac{\sum_{i=q}^{\infty}\left(\frac{A}{n}\right)^i \cdot \frac{A^n}{n!}}{\sum_{i=o}^{n-1}\frac{A^i}{i!} + \frac{n}{n-A}\cdot\frac{A^n}{n!}}, x \geq n$$

$$P_{x\geq n+q} = \frac{\left(\frac{A}{n}\right)^q \cdot \frac{A^n}{n!} \cdot \sum_{i=0}^{\infty}\left(\frac{A}{n}\right)^i}{\sum_{i=o}^{n-1}\frac{A^i}{i!} + \frac{n}{n-A}\cdot\frac{A^n}{n!}} = \frac{\left(\frac{A}{n}\right)^q \cdot \frac{A^n}{n!} \cdot \frac{n}{n-A}}{\sum_{i=o}^{n-1}\frac{A^i}{i!} + \frac{n}{n-A}\cdot\frac{A^n}{n!}}, x \geq n$$

$$P_{x\geq n+q} = D_n(A).\left(\frac{A}{n}\right)^q, x \geq n$$

3. Erlang's Delay System (M/M/n - System)

Mean queue length E{q}:

$$E\{q\} = \sum_{q=0}^{\infty} q.P_{n+q} = \frac{\sum_{q=0}^{\infty} q.\left(\frac{A}{n}\right)^{q} \cdot \frac{A^n}{n!}}{\sum_{i=0}^{n-1} \frac{A^i}{i!} + \frac{n}{n-A} \cdot \frac{A^n}{n!}} = \frac{\frac{A^n}{n!} \cdot \sum_{q=0}^{\infty} q.\left(\frac{A}{n}\right)^{q}}{\sum_{i=0}^{n-1} \frac{A^i}{i!} + \frac{n}{n-A} \cdot \frac{A^n}{n!}}$$

$$\sum_{q=0}^{\infty} q.\left(\frac{A}{n}\right)^{q} = \frac{A}{n} \cdot \frac{\partial}{\partial\left(\frac{A}{n}\right)} \left\{ \sum_{q=0}^{\infty} \left(\frac{A}{n}\right)^{q} \right\} = \frac{A}{n} \cdot \frac{\partial}{\partial\left(\frac{A}{n}\right)} \left\{ \frac{1}{1-\frac{A}{n}} \right\} = \frac{A/n}{\left(1-\frac{A}{n}\right)^{2}}$$

$$E\{q\} = \frac{\frac{A^n}{n!} \cdot \frac{A}{n} \cdot \left(\frac{n-A}{n}\right)^{-2}}{\sum_{i=0}^{n-1} \frac{A^i}{i!} + \frac{n}{n-A} \cdot \frac{A^n}{n!}} = \frac{A}{n-A} \cdot \frac{\frac{A^n}{n!} \cdot \frac{n}{n-A}}{\sum_{i=0}^{n-1} \frac{A^i}{i!} + \frac{n}{n-A} \cdot \frac{A^n}{n!}}$$

$$E\{q\} = D_n(A).\frac{A}{n-A}$$

3. Erlang's Delay System (M/M/n - System)

Little' s law : $E\{q\} = \lambda . t_{w,all}$

$$t_{w,all} = \frac{1}{\lambda} . D_n(A) . \frac{A}{n-A} = D_n(A) . \frac{A.t_m}{n-A}$$

Average number of customers in the system $E\{x\}$

$$E\{x\} = A + E\{q\} = A + D_n(A) . \frac{A}{n-A}$$

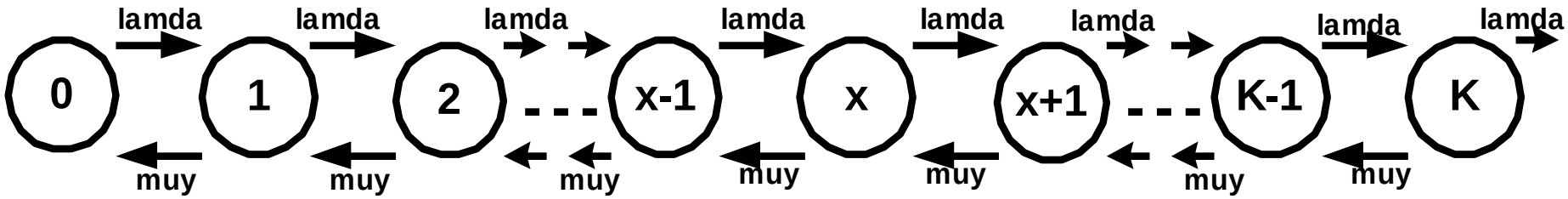
Average time spent in the system $E\{t\}$:

$$E\{t\} = t_m + t_{w,all} = t_m + D_n(A) . \frac{A.t_m}{n-A}$$

$Utilization : \rho = A/n$, average carried load per server.

4. Erlang's Loss Delay System (M/M/1/K - System)

The state transition diagram:



In statistical equilibriu m :

$$\lambda.P_0 = \mu.P_1 \rightarrow P_1 = \frac{\lambda}{\mu}.P_0$$

$$\lambda.P_1 = \mu.P_2 \rightarrow P_2 = \left(\frac{\lambda}{\mu}\right)^2.P_0 = A^2.P_0$$

$$\lambda.P_{x-1} = \mu.P_x \rightarrow P_x = A^x.P_0$$

Normalization :

$$1 = \sum_{x=0}^{K} P_x = P_0.\sum_{x=0}^{K} A^x = P_0.\frac{1-A^{K+1}}{1-A} \rightarrow \left\{ \begin{array}{l} P_0 = \dfrac{1-A}{1-A^{K+1}} \\ P_x = \dfrac{A^x(1-A)}{1-A^{K+1}} \end{array} \right\}$$

$$\text{Blocking Probability } \mathrm{B_c} = \mathrm{B_T} = \mathrm{P_K} = \frac{A^K(1-A)}{1-A^{K+1}}$$

4. Erlang's Loss Delay System (M/M/1/K - System)

Dimentioni ng M/M/1/K system:

Assume that B_c and A are given. The task is to solve minimum size K so that $B_c \leq B$.

$$B_c = \frac{A^K(1-A)}{1-A^{K+1}} \leq B \rightarrow A^K - A^{K+1} = B - B.A^{K+1}$$

$$A^K = \frac{B}{1-A.(1-B)} \rightarrow K = \frac{\log\left(\frac{B}{1-A.(1-B)}\right)}{\log(A)}$$

$$P\{\text{being served immediatel y}\} = P\{S\} = \sum_{x=0}^{0} P_x$$

$$P\{\text{to enter the queue}\} = P\{W\} = \sum_{x=1}^{K-1} P_x$$

$$P\{\text{to be lost}\} = P\{B\} = P_K$$

4. Erlang's Loss Delay System (M/M/1/K - System)

Average queue length in M/M/1/K :

$$P_x = \frac{A^x(1-A)}{1-A^{K+1}}, \text{ x} = 0,1,\ldots,\text{K and A} = \lambda/\mu$$

$$\text{E}\{\text{q}\} = \sum_{x=2}^{K}(x-1).P_x = \sum_{q=2}^{K}(q-1).\frac{A^q(1-A)}{1-A^{K+1}}$$

$$E\{q\} = \frac{A^2(1-A)}{1-A^{K+1}}.\sum_{q=2}^{K}(q-1).A^{q-1} = \frac{A^2(1-A)}{1-A^{K+1}}.\sum_{q=1}^{K-1}q.A^{q-1}$$

$$E\{q\} = \frac{A^2(1-A)}{1-A^{K+1}}.\sum_{q=1}^{K-1}\frac{\partial}{\partial A}\left(A^q\right) = \frac{A^2(1-A)}{1-A^{K+1}}.\frac{\partial}{\partial A}\left(\frac{A-A^K}{1-A}\right)$$

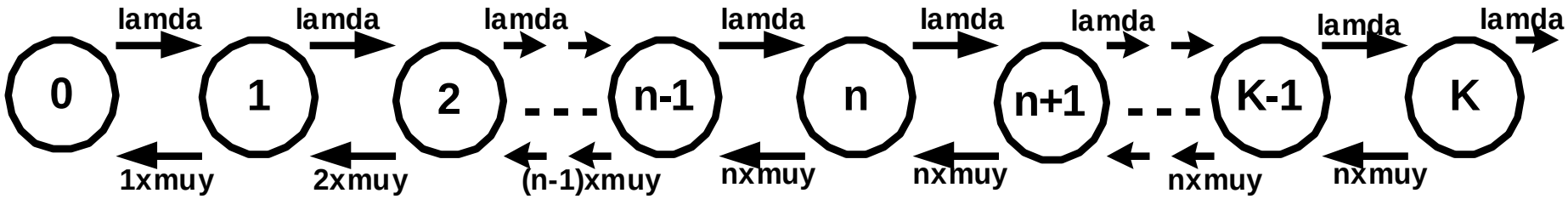
$$E\{q\} = \frac{A^2(1-A)}{1-A^{K+1}}.\frac{1-A^{K-1}.[K(1-A)+A]}{(1-A)^2}$$

$$E\{q\} = \frac{A^2\{1-A^{K-1}.[K(1-A)+A]\}}{(1-A).(1-A^{K+1})}$$

Mean waiti ng time is (using Little' s law) : $t_{w,all} = E\{q\}/\lambda$

4. Erlang's Loss Delay System (M/M/n/K - System)

The state transition diagram:



In statistical equilibriu m :

$$\lambda.P_0 = \mu.P_1 \rightarrow P_1 = (\lambda/\mu).P_0 = A.P_0$$

$$\lambda.P_1 = 2\mu.P_2 \rightarrow P_2 = (\lambda/\mu)^2/2.P_0 = \frac{A^2}{2!}.P_0$$

$$\lambda.P_{n-1} = n\mu.P_n \rightarrow P_n = \frac{A^n}{n!}.P_0$$

$$\lambda.P_n = n\mu.P_{n+1} \rightarrow P_{n+1} = \frac{A}{n}.\frac{A^n}{n!}.P_0$$

$$\lambda.P_{K-1} = n\mu.P_K \rightarrow P_K = \left(\frac{A}{n}\right)^{K-n}.\frac{A^n}{n!}.P_0$$

4. Erlang's Loss Delay System (M/M/n/K - System)

$$1=\sum_{x=0}^{K}P_x=P_0.\left[\sum_{x=0}^{n-1}\frac{A^x}{x!}+\sum_{q=0}^{K-n}\left(\frac{A}{n}\right)^q.\frac{A^n}{n!}\right]=\left\{\begin{array}{l} P_0.\left[\sum_{x=0}^{n-1}\frac{A^x}{x!}+\frac{A^n}{n!}.\frac{1-(A/n)^{K-n+1}}{1-(A/n)}\right]\text{for (A/n)}\neq 1 \\ P_0.\left[\sum_{x=0}^{n-1}\frac{A^x}{x!}+\frac{A^n}{n!}.(K-n+1)\right]\text{for (A/n)}=1 \end{array}\right\}$$

$$P_{x=0,1,\ldots,n-1}=\frac{\frac{A^x}{x!}}{\left[\sum_{x=0}^{n-1}\frac{A^x}{x!}+\frac{A^n}{n!}.\frac{1-(A/n)^{K-n+1}}{1-(A/n)}\right]}=\left[\frac{\frac{A^x}{x!}}{\left[\sum_{x=0}^{n-1}\frac{A^x}{x!}+\frac{A^n}{n!}.(K-n+1)\right]}\right]_{(A/n)=1}$$

$$P_{\substack{x=n,\ldots,K\\ q=0,\ldots,K-n}}=\frac{\left(\frac{A}{n}\right)^q.\frac{A^n}{n!}}{\left[\sum_{x=0}^{n-1}\frac{A^x}{x!}+\frac{A^n}{n!}.\frac{1-(A/n)^{K-n+1}}{1-(A/n)}\right]}=\left[\frac{\frac{A^n}{n!}}{\left[\sum_{x=0}^{n-1}\frac{A^x}{x!}+\frac{A^n}{n!}.(K-n+1)\right]}\right]_{(A/n)=1}$$

4. Erlang's Loss Delay System (M/M/n/K - System)

$$P\{\text{immediate service}\} = \sum_{x=0}^{n-1} P_x = \frac{\sum_{x=0}^{n-1} \frac{A^x}{x!}}{\sum_{i=0}^{n-1} \frac{A^i}{i!} + \frac{A^n}{n!} \cdot \frac{1-(A/n)^{K-n+1}}{1-(A/n)}}$$

$$P\{\text{delayed service}\} = \sum_{x=n}^{K-1} P_x = \sum_{q=0}^{K-n-1} P_{x=n+q} = \frac{\frac{A^n}{n!} \cdot \frac{1-(A/n)^{K-n}}{1-(A/n)}}{\sum_{i=0}^{n-1} \frac{A^i}{i!} + \frac{A^n}{n!} \cdot \frac{1-(A/n)^{K-n+1}}{1-(A/n)}}$$

$$P\{\text{loss}\} = P_k = \frac{\left(\frac{A}{n}\right)^{K-n} \cdot \frac{A^n}{n!}}{\sum_{i=0}^{n-1} \frac{A^i}{i!} + \frac{A^n}{n!} \cdot \frac{1-(A/n)^{K-n+1}}{1-(A/n)}}$$

4. Erlang's Loss Delay System (M/M/n/K - System)

Average queue length in M/M/n/K :

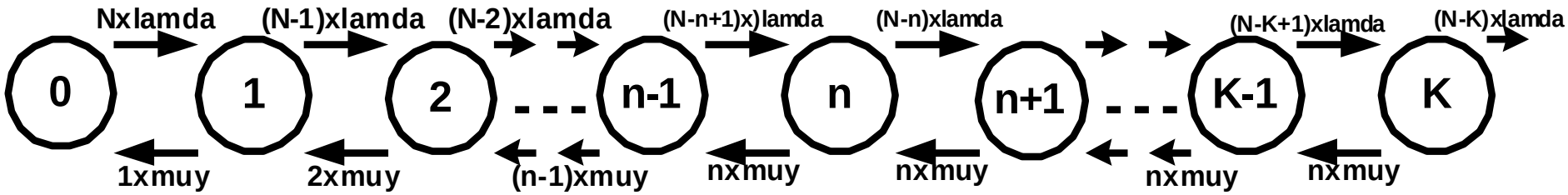
$$E\{q\} = \sum_{q=1}^{K-n} q.P_{n+q} = \sum_{q=1}^{K-n} q.(A/n)^q.P_n = (A/n).P_n.\sum_{q=1}^{K-n} q.(A/n)^{q-1} = \left[\sum_{q=1}^{K-n} q.P_n\right]_{(A/n)=1}$$

$$E\{q\} = (A/n).P_n.\sum_{q=1}^{K-n} \frac{\partial}{\partial(A/n)}.(A/n)^q = (A/n).P_n.\left[\frac{\partial}{\partial(A/n)}\left(\frac{(A/n)-(A/n)^{K-n+1}}{1-(A/n)}\right)\right]_{(A/n)\neq 1}$$

$$E\{q\} = \left\{ \begin{array}{l} \dfrac{(A/n).P_n}{[1-(A/n)]^2}.\left\{\left[1-\left(\dfrac{A}{n}\right)^{K-n}\right].\left[1+(K-n).(1-\dfrac{A}{n})\right]\right\}_{(A/n)\neq 1} \\ \left\{\dfrac{(K-n).(K-n+1).\dfrac{A^n}{n!}}{2.\left[\displaystyle\sum_{i=0}^{n-1}\dfrac{A^i}{i!}+\dfrac{A^n}{n!}.(K-n+1)\right]}\right\}_{(A/n)=1} \end{array} \right\}$$

4. Erlang's Loss Delay System (M/M/n/K/N - System)

The state transition diagram:



In statistical equilibriu m :

$$N\lambda.P_0 = \mu.P_1 \rightarrow P_1 = N(\lambda/\mu).P_0 = Na.P_0$$

$$(N-1)\lambda.P_1 = 2\mu.P_2 \rightarrow P_2 = N(N-1)\frac{a^2}{2!}.P_0 = \frac{N!}{(N-2)!}.\frac{a^2}{2!}.P_0$$

$$(N-n+1)\lambda.P_{n-1} = n\mu.P_n \rightarrow P_n = N(N-1)\ldots(N-n+1)\frac{a^n}{n!}.P_0 = \frac{N!}{(N-n)!}.\frac{a^2}{2!}.P_0$$

$$(N-n)\lambda.P_n = n\mu.P_{n+1} \rightarrow P_{n+1} = N(N-1)\ldots(N-n)\frac{a}{n}.\frac{a^n}{n!}.P_0$$

$$(N-K+1)\lambda.P_{K-1} = n\mu.P_K \rightarrow P_K = N(N-1)\ldots(N-K+1)\left(\frac{a}{n}\right)^{K-n}.\frac{a^n}{n!}.P_0$$

4. Erlang's Loss Delay System (M/M/n/K/N - System)

$$P_x = \frac{N!}{(N-x)!}.\frac{a^x}{x!}.P_0, x = 0,1,\ldots,n$$

$$P_x = \frac{(N-n)!}{(N-x)!}.\left(\frac{a}{n}\right)^{x-n}.P_n, x = n,\ldots,K$$

$$1 = \sum_{x=0}^{\infty} P_x = P_0.\left[\sum_{x=0}^{n-1}\frac{N!}{(N-x)!}.\frac{a^x}{x!} + \sum_{x=n}^{K}\frac{(N-n)!}{(N-x)!}.\left(\frac{a}{n}\right)^{x-n}.\frac{N!}{(N-n)!}.\frac{a^n}{n!}\right]$$

$$P_x = \frac{\dfrac{N!}{(N-x)!}.\dfrac{a^x}{x!}}{\left[\sum_{x=0}^{n-1}\dfrac{N!}{(N-x)!}.\dfrac{a^x}{x!} + \sum_{x=n}^{K}\dfrac{(N-n)!}{(N-x)!}.\left(\dfrac{a}{n}\right)^{x-n}.\dfrac{N!}{(N-n)!}.\dfrac{a^n}{n!}\right]}, x = 0,1,\ldots,n-1$$

$$P_x = \frac{\dfrac{(N-n)!}{(N-x)!}.\left(\dfrac{a}{n}\right)^{x-n}.\dfrac{N!}{(N-n)!}.\dfrac{a^n}{n!}}{\left[\sum_{x=0}^{n-1}\dfrac{N!}{(N-x)!}.\dfrac{a^x}{x!} + \sum_{x=n}^{K}\dfrac{(N-n)!}{(N-x)!}.\left(\dfrac{a}{n}\right)^{x-n}.\dfrac{N!}{(N-n)!}.\dfrac{a^n}{n!}\right]}, x = n,\ldots,K$$

The total arrival rate is : $\lambda_T = \sum_{x=0}^{K}\lambda_x.P_x$

4. Erlang's Loss Delay System (M/M/n/K/N - System)

$$P\{\text{immediate service}\} = \sum_{x=0}^{n-1} \frac{\lambda_x . P_x}{\lambda_T}$$

$$P\{\text{delayed service}\} = \sum_{x=n}^{K-1} \frac{\lambda_x . P_x}{\lambda_T}$$

$$P\{\text{loss}\} = \frac{\lambda_K . P_K}{\lambda_T}$$

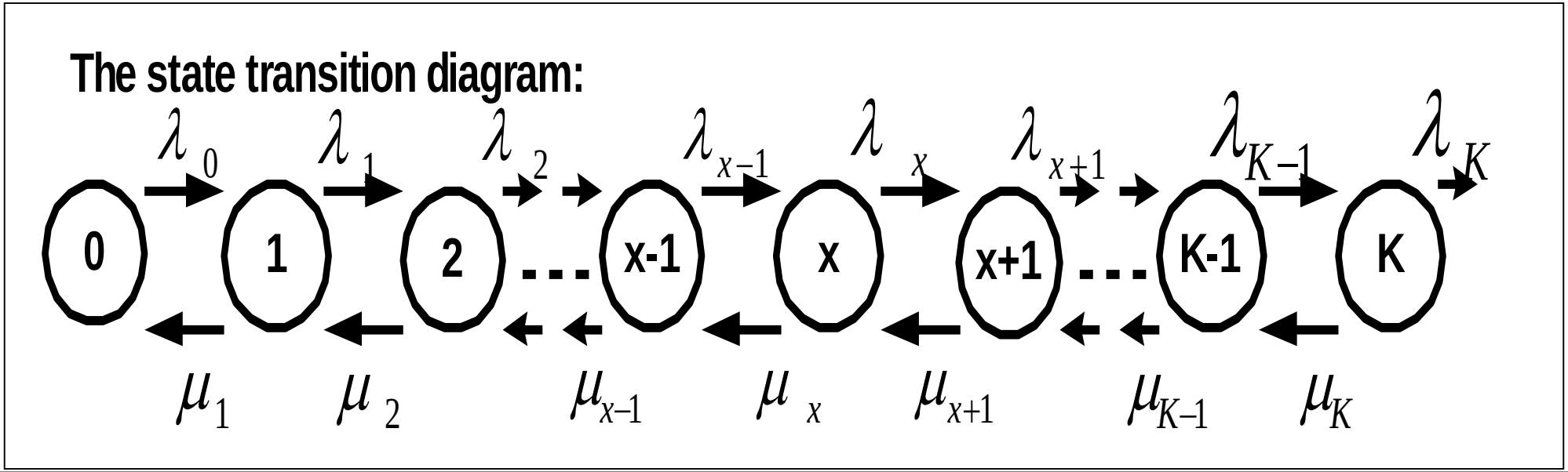
4. General birth and death process (M/M/n/K/N-FIFO)

✓Both inter-arrival times and service times are exponentially distributed.

✓The state x(t) of the system at time t is the number of customers staying in the system at this time (waiting or being served)

✓Both the arrival intensity and the service rate are dependent on the state of the system

✓**<u>The state transition diagram is below</u>**



4. General birth and death process (M/M/n/K/N-FIFO)

In statistical equilibrium (node equation) :

$$0 = \lambda_{x-1}.P(x-1) - (\lambda_x + \mu_x).P(x) + \mu_{x+1}.P(x+1)$$

$$= -\lambda_x.P(0) + \mu_1.P(1)$$

All state probabilities P(x) can be expressed by

$$\mathrm{P(x)} = \frac{\lambda_0.\lambda_1....\lambda_{x-1}}{\mu_1.\mu_2...\mu_x}.P(0)$$

P(0) is solved by normalizing $\sum \mathrm{P(x)} = 1$

4. General birth and death process (M/M/n/K/N-FIFO)

$$\text{Mean queue length (All states)}: \mathrm{E}\{\mathrm{q}\} = \sum_{\mathrm{x=n+1}}^{\mathrm{K}} (x-n).P(x)$$

$$\text{Mean queue length, given the queue exists } (\mathrm{x} < \mathrm{n}):$$

$$\mathrm{E}\{\mathrm{q} \mid \mathrm{x} > \mathrm{n}\} = \frac{\mathrm{E}\{\mathrm{q}\}}{\sum_{\mathrm{x=n+1}}^{\mathrm{K}} P(x)}$$

$$\text{Total arrival rate}: \lambda_{\mathrm{T}} = \sum_{\mathrm{x=0}}^{\mathrm{K}} \lambda_x.P(x)$$

4. General birth and death process (M/M/n/K/N-FIFO)

$$P\{\text{being served immediatel y}\} = P\{S\}$$

$$P\{S\} = \sum_{x=0}^{n-1} \lambda_x . P(x) / \lambda_T$$

$$P\{\text{to enter the queue}\} = P\{W\}$$

$$P\{W\} = \sum_{x=n}^{K-1} \lambda_x . P(x) / \lambda_T$$

$$P\{\text{to be lost}\} = P\{B\}$$

$$P\{S\} = \lambda_K . P(K) / \lambda_T$$

5. Little's Formula

Let

$P_n(t)$: Probability of n customers waiting in queue or under service at time t and:

$$\overline{N}(t) = \sum_{n=0}^{\infty} nP_n(t)$$

is the average number in the system at time t, we have:

$$\lim_{t \to \infty} P_n(t) = P_n, n = 0,1,\ldots$$

$$N = \sum_{n=0}^{\infty} nP_n = \lim_{t-\infty} \overline{N}(t)$$

Where N : Average number in the system.

5. Little's Formula

$$T = \lim_{t-\infty} \overline{T}_k = \lim_{t-\infty} \frac{1}{k} \sum_{i=1}^{k} T_i, \text{T : Average delay}$$

and $\overline{\mathrm{T}}_{\mathrm{k}}$: Average delay of the kth customer, typically converges as $\mathrm{k} \to \infty$ to a steady-state value

T_{i} : the delay of the ith customer

$\Rightarrow$ Little Formula : $\quad \mathrm{N} = \lambda.\mathrm{T}$

λ : Average customer arrival rate, is given by :

$$\lambda = \lim_{t \to \infty} \frac{\text{Expected number of arrivals in the interval } [0, \mathrm{t}]}{t}$$

6. General Results in Queuing Theory

Little's theorem is the most general result which is valid for an arbitrary queuing system. The theorem is easy to apply and very useful in many cases.

Only queuing systems with Poisson arrival processes are simple to deal with.

Erlang's Delay Formula is valid only for Poisson input ($N=\infty$,exponential inter-arrival times, mean $1/\lambda$), and only for exponential service time distribution.

Erlang's Loss Formula (Poisson input) is valid for arbitrary holding time distribution for any n.